

Day3:

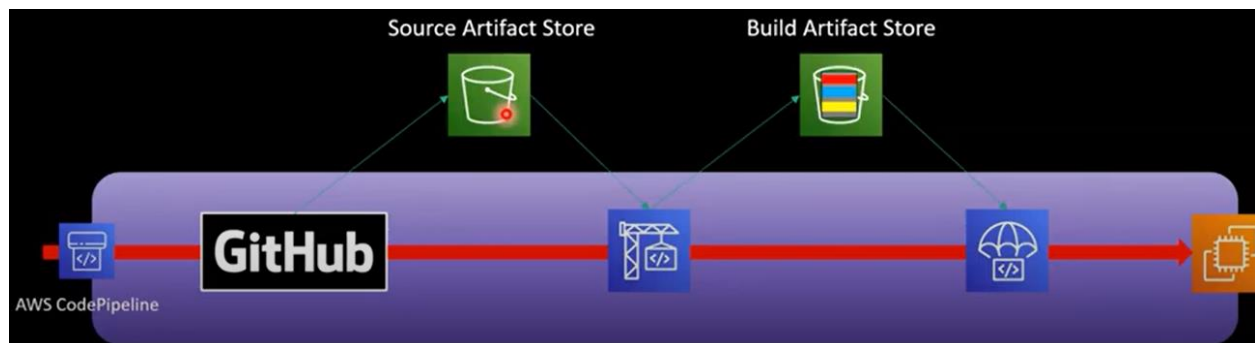
CodeDeploy

- Using CodeDeploy we deploy the changes in production server, It is AWS managed service to automate your deployment, result in faster deployment.
- Rollback update feature
- Blue-green deployment

Scalable.... It can be deployed to one or thousands of servers.

Deployment can be done on below

- On-premises servers
- EC2 Instances
- AWS Lambda
- Amazon ECS



step1: As a developer handover the raw code (index.html), appspec. yaml and script folder.

version: 0.0

os: linux

files:

- source: /
destination: /home/ubuntu

hooks:

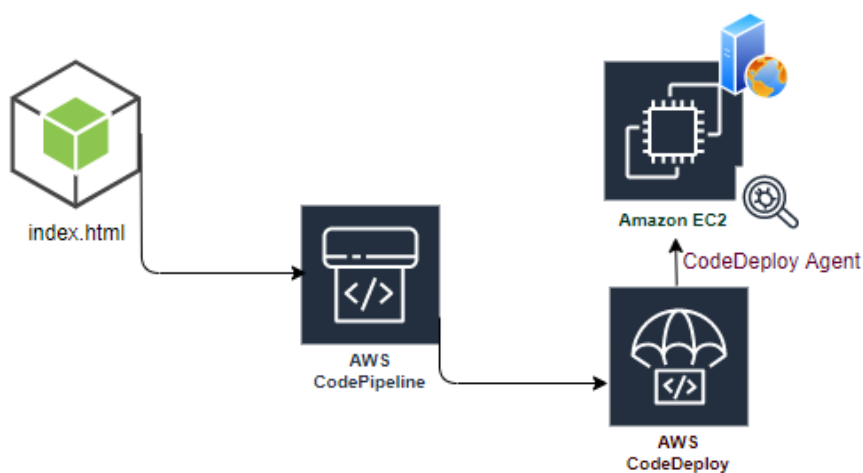
Afterinstall:

- location: after.sh
timeout: 100

Applicationstart:

- location: applicationstart.sh
timeout: 100

step2: Whenever there is a change deploy it in to the EC2 server.



Continuous Integration happens through the **CodePipeline**, **code deployment** happens through the **CodeDeploy**.

Whenever there is a change in index.html, change, pipeline notified, changes picked up by CodePipeline, and request CodeDeploy service, hey! please put this file in specific folder in EC2 server.

In EC2 server, CodeDeploy agent already installed, this CodeDeploy agent gets the executable from S3 (its different default specific S3 bucket).

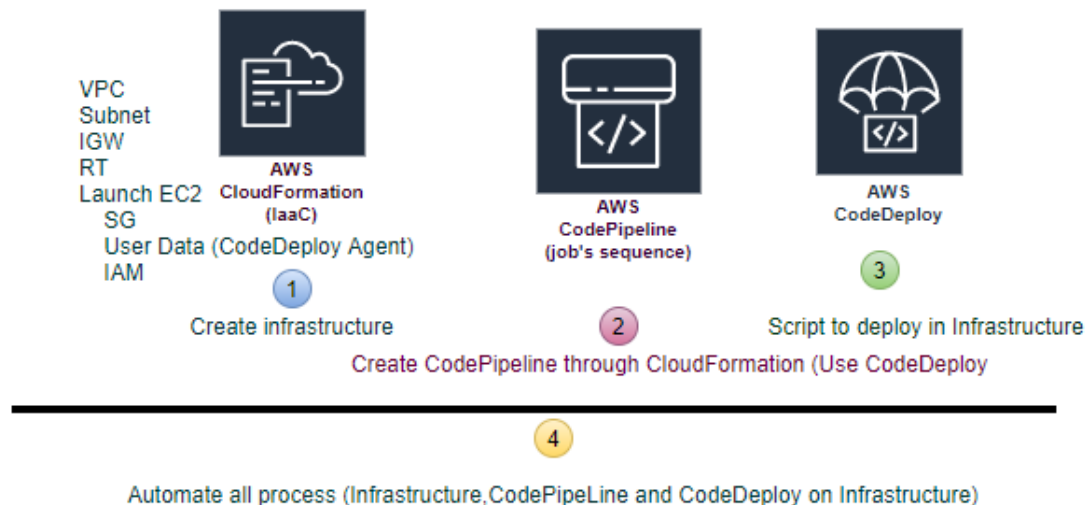
EC2 as a service has to communicate with S3 (EC2 must have role to interact with S3)

Policy: AWSEC2RoleforAWSCodeDeploy

Trust Relationship: ec2

- handover the piece of code to CodePipeline, CodePipeline handover to the CodeDeploy, CodeDeploy communicates with agent, and this agent communicate back to CodeDeploy Service (443).
- EC2 security group must carry TCP 443.
- There may be number of EC2, called deployment group, each of them must have (CodeDeploy agent).

What is CloudFormation?



UC: CodeDeploy is input for CodePipeline, first create a CodeDeploy

step1: CodeDeploy-> deploy->application-> Create Application

step2: Create Deployment Group (fleet of EC2 instances)

Create CodePipeline

step3: Pipelines-> Create Pipeline

step4: Add Source (In case of S3 version must be enabled on S3 bucket)

step5: Build Provider (Optional in case of static contents)

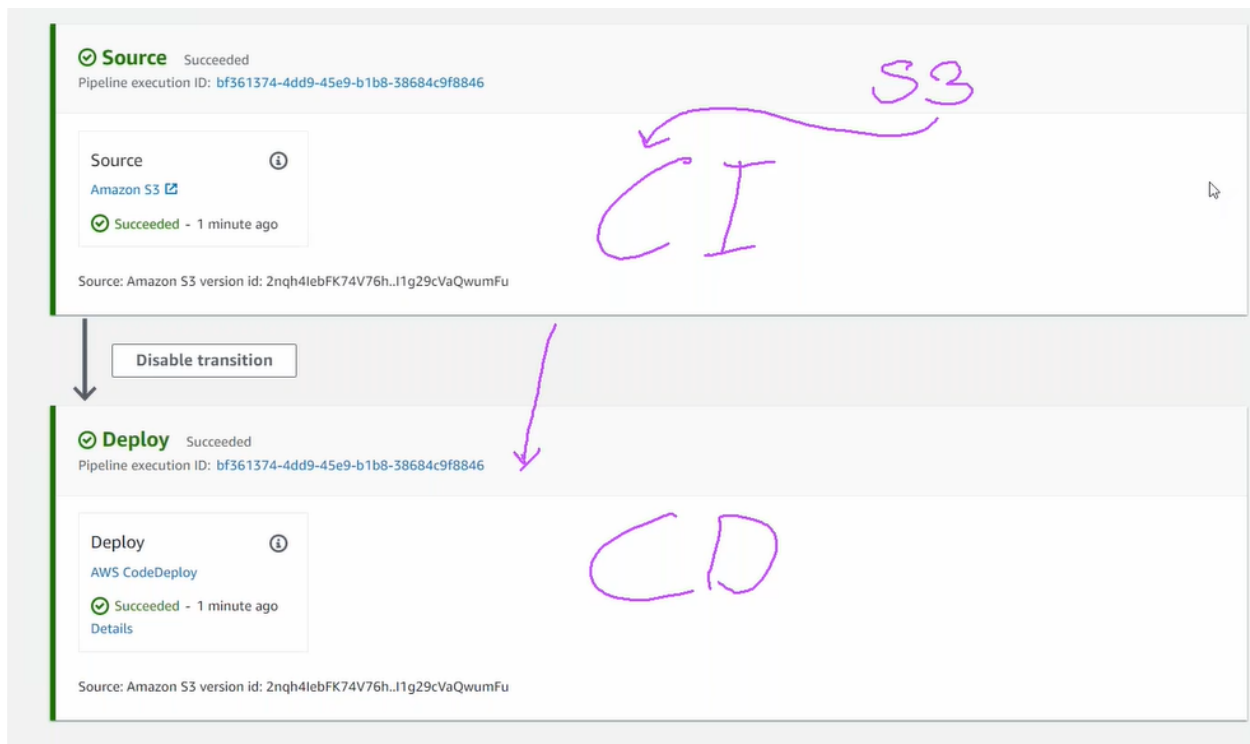
step6: Deploy Provider (CodeDeploy).

step7: CodeDeploy Service communicated with the agent on the EC2 Instance.

Events (hooks): Pipeline->Deploy->Details

Event	Duration	Status	Error code	Start time	End time
ApplicationStop	less than one second	✓ Succeeded	-	Jun 19, 2021 7:41 AM (UTC+5:30)	Jun 19, 2021 7:41 AM (UTC+5:30)
DownloadBundle	2 seconds	✓ Succeeded	-	Jun 19, 2021 7:41 AM (UTC+5:30)	Jun 19, 2021 7:42 AM (UTC+5:30)
BeforeInstall	1 minute 49 seconds	✓ Succeeded	-	Jun 19, 2021 7:42 AM (UTC+5:30)	Jun 19, 2021 7:43 AM (UTC+5:30)
Install	less than one second	✓ Succeeded	-	Jun 19, 2021 7:43 AM (UTC+5:30)	Jun 19, 2021 7:43 AM (UTC+5:30)
AfterInstall	less than one second	✓ Succeeded	-	Jun 19, 2021 7:43 AM (UTC+5:30)	Jun 19, 2021 7:43 AM (UTC+5:30)
ApplicationStart	less than one second	✓ Succeeded	-	Jun 19, 2021 7:43 AM (UTC+5:30)	Jun 19, 2021 7:43 AM (UTC+5:30)
ValidateService	less than one second	✓ Succeeded	-	Jun 19, 2021 7:43 AM (UTC+5:30)	Jun 19, 2021 7:43 AM (UTC+5:30)

If there are some changes in S3 object, it notified to Code Pipeline. Code Pipeline immediate fetch the latest artifact and handover it to code deploys.



step8: How handover happens? Through S3

CodeDeploy handover the complete .zip file to CodeDeploy Agent, .zip file contains appspec.yaml: has all the necessary instructions for CodeDeploy Agent to follow **and** complete the deployment process. CodeDeploy service just handover the .zip to CodeDeploy Agent, Its CodeDeploy Agent actually do the deployment process.

```
version: 0.0
os: windows
files:
  - source: \index.html
    destination: C:\inetpub\wwwroot
hooks:
  BeforeInstall:
    - location: \before-install.bat
      timeout: 900
```

CI: source code-> CodePipeline

How S3 communicate with CodePipeline, that there is some change?

CloudWatch, has Events ->(Rule) defined, created by CodePipeline (Policy and Service Rule Created by CodePipeline).

CloudTrail, keeps eye on each and every activity that are happening in your account.

Rule says hey! CloudTrail, keep eye on S3 object, if S3 event (code change) triggered, then please trigger the event (rule). Trigger says, call the CodePipeline.

Integration is series of services connected to each other, S3 has event triggered when change the version, the event was picked up by CloudTrail, CloudTrail notified CloudWatch, and CloudWatch trigger the Pipeline.

CodeBuild (buildspec.yaml):

version: 0.2

phases:

build:

commands:

- echo "This is built phase"

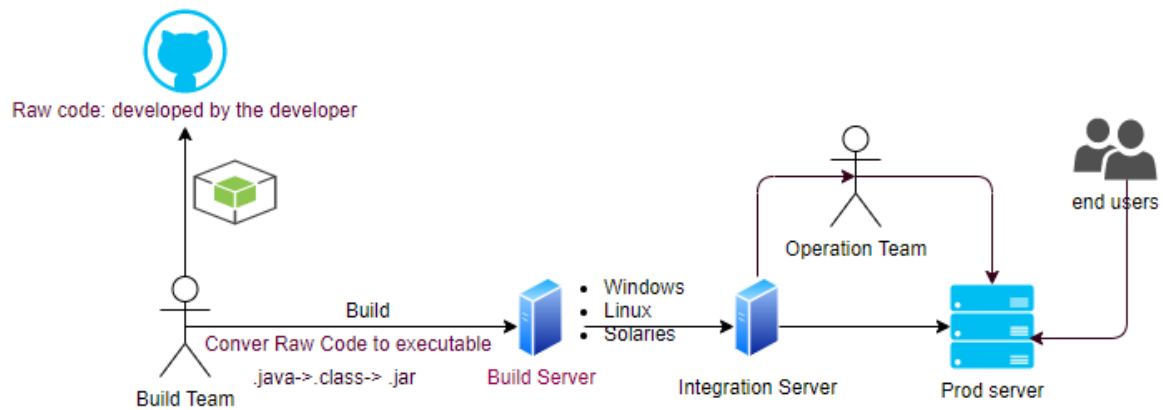
post_build:

commands:

- echo "Deploy to S3"

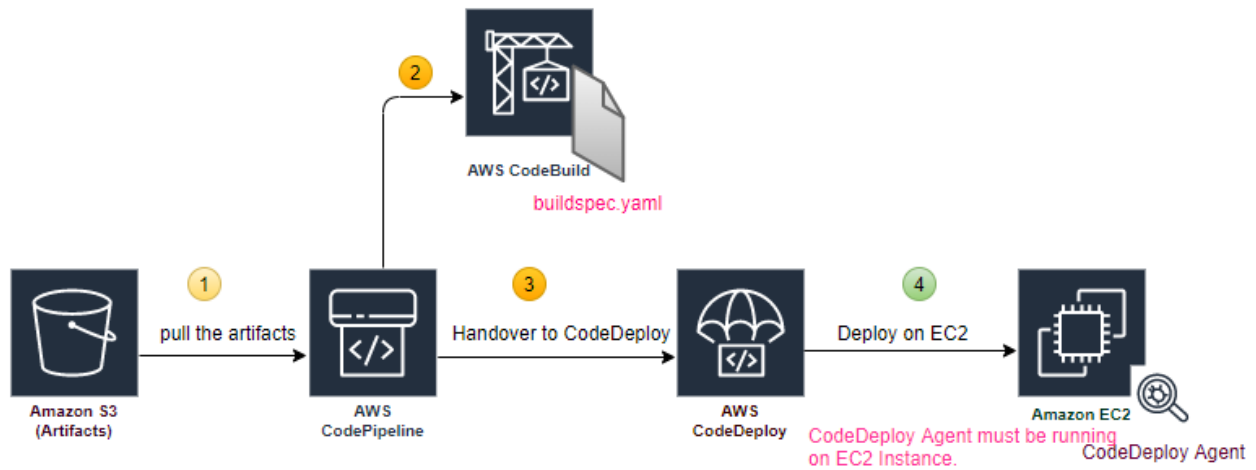
How to get Raw file and converts to the artifacts.

Just like maven, process of converting raw code in to executable (artifacts) e.g. .java->.class. Generates the /target



```
1 version: 0.2
2
3 phases:
4   install:
5     runtime-versions:
6       java: corretto8
7   build:
8     commands:
9       - echo Build started on `date`
10      - mvn package
11   post_build:
12     commands:
13       - echo Build completed on `date`
14       - mvn test
15 artifacts:
16   files:
17     - target/my-app-1.0-SNAPSHOT.jar
18     - appspec.yml
19   discard-paths: yes
```

mvn package

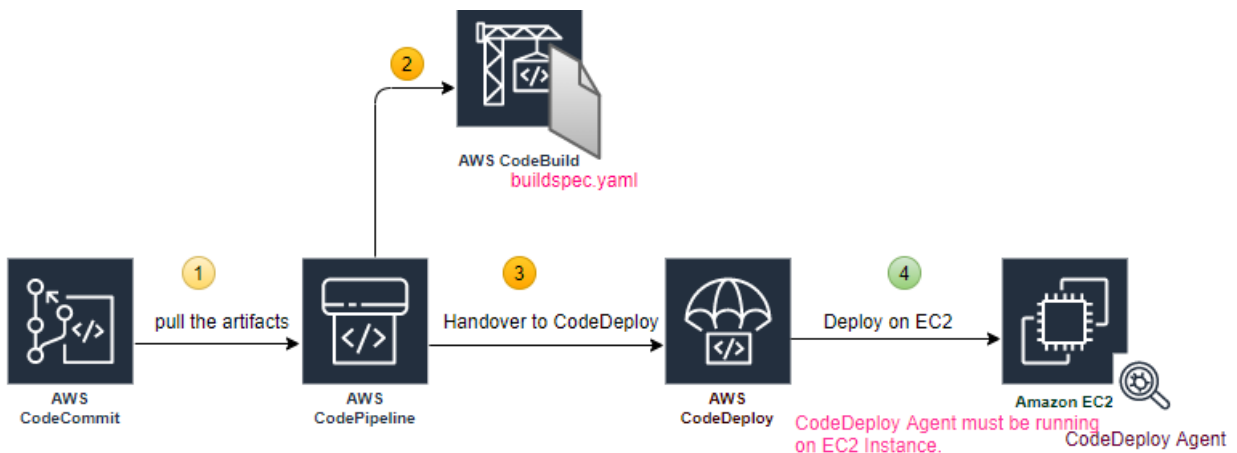


Artifacts:

CodeArtifacts (Rather than uploading in S3, upload the artifacts in CodeArtifacts).

CodeCommit (multiple users working on same artifacts):

Just like GitHub, remote repository (Underneath uses Git).



step1: CodeCommit-> Repository-> Create Repository

Create repository

Create a secure repository to store and share your code. Begin by typing a repository name and a description for your repository. Repository names are included in the URLs for that repository.

Repository settings

Repository name

100 characters maximum. Other limits apply.

Description - *optional*

1,000 characters maximum

Tags

Add

Cancel

Create

Developer Tools

CodeCommit

▼ Source • CodeCommit

Getting started

Repositories

Approval rule templates

► Artifacts • CodeArtifact

Developer Tools > CodeCommit > Repositories

Repositories

Info

🔄

🔔 Notify ▼

📄 Clone URL ▼

👁 View repository

🗑 Delete repository

Create repository

< 1 >

⚙

	Name	Description	Last modified	Clone URL
☐	TestRepo	Testing purpose only	Just now	🔗 HTTPS 🔗 SSH 🔗 HTTPS (GRC)

step2: In command line create folder

step2.1: git init

step2.2: git clone <url>

step3: If Using CodeCommit first time.

- IAM User-> Security Credentials-> Generate Git Credentials
- User must have Policy **AWSCodeCommitPowerUser**
- Have access to the Repository

No results

HTTPS Git credentials for AWS CodeCommit

Generate a user name and password you can use to authenticate HTTPS connections to AWS CodeCommit repositories. You can generate and store up to 2 sets of credentials. [Learn more](#)

Generate credentials

No credentials have been generated.

Credentials for Amazon Keyspaces (for Apache Cassandra)

Generate a user name and password you can use to authenticate to Amazon Keyspaces. You can generate and store up to 2 sets of credentials for Amazon Keyspaces. [Learn more](#)

Generate credentials

2:45 stuck